

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 12: Reflection

Every formula and result of the chapter on one sheet

1	Reflection in the x-axis	$(x, y) \rightarrow (x, -y)$
2	Reflection in the y-axis	$(x, y) \rightarrow (-x, y)$
3	Reflection in the origin	$(x, y) \rightarrow (-x, -y)$
4	Reflection in the line $x = a$	$(x, y) \rightarrow (2a - x, y)$
5	Reflection in the line $y = b$	$(x, y) \rightarrow (x, 2b - y)$
6	Mirror line	Perpendicular bisector of the segment joining a point and its image
7	Invariant point	A point on the mirror line; its image is itself
8	M_x followed by M_y	Same as reflection in the origin
9	Equation of the x-axis	$y = 0$
10	Equation of the y-axis	$x = 0$
11	Figure and its image	Congruent, with lengths and angles unchanged